HireX — Product Requirements Document (PRD)

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Target Audience: Product Managers, Business Analysts, QA Leads, Software Developers, Stakeholders
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1. Document Purpose, Version & Scope

1.1 Purpose

This Product Requirements Document (PRD) provides the definitive specification of the HireX recruitment platform. It captures verified business workflows, user journeys, functional and non-functional requirements, acceptance criteria, and system guardrails as implemented in the HireX codebase.

1.2 Scope

- **Candidate Portal:** Branded career pages, job search, dynamic multi-step application form wizard with autosave, interview slot self-booking, electronic offer review, and onboarding document uploads.
- **Recruiter Dashboard:** Tenant organization setup, job requisition with custom form builder, Kanban pipeline state machine, AI resume scoring (`aiScore`), interview scheduling, offer generation, and document verification.
- **Interviewer Workspace:** Queue of assigned candidate interviews, recurring weekly availability calendar, candidate resume review, private recruiter briefings, and rubric scorecards with Cultural Fit rating.
- **Backend Services:** Better-Auth session cookie partitioning, Schedular (MeetX) video room lifecycle, Cloudflare R2 multi-file storage, and job distribution XML feeds.

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2. Product Overview & Problem Statement

2.1 The Business Problem

Modern recruitment teams struggle with fragmented software stacks:

- **High Candidate Drop-Off:** Rigid, repetitive application forms without draft auto-save cause candidates to abandon applications mid-way.
- **Scheduling Friction:** 4 to 6 back-and-forth emails between recruiters, candidates, and interviewers to coordinate a single 45-minute technical session.
- **Pipeline Data Leakage:** Lack of forward-only state progression leads to accidental reversions of candidates who have already passed technical rounds.

- **Post-Offer Drop-Off & Compliance Failures:** Disconnected document collection via unencrypted email attachments leads to missing compliance documents prior to joining.

2.2 The HireX Solution

HireX provides an integrated, multi-tenant talent operating system:

1. **Dynamic Form Builder & Wizard:** Custom questions per job posting with local draft autosave and `beforeunload` guards.
2. **MeetX Real-Time Video Orchestration:** Self-service slot booking backed by recurring interviewer availability and automated WebRTC room links.
3. **Forward-Only State Machine:** Strict validation rules preventing invalid regressions (e.g. no regression to `APPLIED`), hard-blocking pipeline advance when interviews remain open.
4. **Compliance Document Verification:** Mandatory `DOCUMENT_VERIFICATION` stage post-offer acceptance with multi-file uploads and in-browser verification.

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3. Stakeholders & User Personas

Persona	Primary Goal	Core Responsibilities	Touchpoint Portal
Candidate (Job Seeker)	Secure employment with transparent updates	Search jobs, fill applications, self-book interview slots, sign offers, upload compliance documents	Candidate Portal (`:3000` or custom CNAME)
Recruiter / HR Admin	Source, evaluate, and hire talent efficiently	Create jobs, customize form questions, manage Kanban pipeline, generate offers, verify documents	Recruiter Portal (`:3002`)
Technical Interviewer	Evaluate candidate technical competence	Configure weekly availability, review candidate resumes and private briefings, submit scorecards	Interviewer Portal (`:3003`)
Hiring Manager	Oversee role requirements and final hiring	Review aggregated scores, approve offers, finalize hires	Recruiter Portal (Scoped access)

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4. Role-Based Access Control (RBAC) Matrix

Action / Capability	Candidate	Recruiter Admin	Recruiter Member	Hiring Manager	Interviewer
Browse Public / Tenant Jobs	✅	✅	✅	✅	✅

Action / Capability	Candidate	Recruiter Admin	Recruiter Member	Hiring Manager	Interviewer
Submit Job Application	✅	❌	❌	❌	❌
Create / Edit Job Postings	❌	✅	✅	❌	❌
Customize Job Application Form	❌	✅	✅	❌	❌
Advance Candidate Pipeline Stage	❌	✅	✅	✅	❌
Schedule Interviews & Assign Panel	❌	✅	✅	❌	❌
View Private Interviewer Briefing	❌	✅	✅	✅	✅
Submit / Update Interview Scorecard	❌	❌	❌	❌	✅
Generate & Extend PDF Offer	❌	✅	❌	❌	❌
Accept / Decline Offer Online	✅	❌	❌	❌	❌
Upload Compliance Documents	✅	❌	❌	❌	❌
Verify / Reject Compliance Documents	❌	✅	✅	❌	❌
Execute "Finalize Hire"	❌	✅	❌	✅	❌
Manage Company Settings & Team	❌	✅	❌	❌	❌

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5. End-to-End Recruitment Lifecycle & State Machine

HireX models the recruitment lifecycle through a forward-only finite state machine:



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6. Functional Requirements (FR) & Acceptance Criteria

Module A: Authentication & Identity

FR-001: Multi-Role Credential Authentication

- **Description:** System must support email and password authentication with distinct role scoping for Candidate, Recruiter, and Interviewer.
- **Linked Test Cases:** `TC-AUTH-001`, `TC-AUTH-002`, `TC-AUTH-004`
- **Acceptance Criteria (Given / When / Then):**
  - **Given** an unauthenticated candidate on the login page,
  - **When** they enter valid registered credentials,
  - **Then** the system establishes a secure HTTP-only session cookie and navigates to the Candidate Dashboard.
  - **When** they enter a password shorter than 8 characters,
  - **Then** the form blocks submission with an inline validation warning.

FR-002: Portal Session Isolation (ADR-001)

- **Description:** Sessions established on one portal must not grant access to another portal with higher privileges.
- **Linked Test Cases:** `TC-AUTH-004`
- **Acceptance Criteria:**
  - **Given** an active Candidate session cookie on `:3000`,

- **When** the browser navigates directly to `:3002/dashboard` (Recruiter) or `:3003/dashboard` (Interviewer),
- **Then** the backend guard returns `403 Forbidden` / `401 Unauthorized` and redirects the user to the portal's `/login` screen.

FR-003: Interviewer Magic Link / Invite Token Redemption

- **Description:** Interviewers can register via single-use invitation tokens generated by recruiters.
- **Linked Test Cases:** `TC-AUTH-003`
- **Acceptance Criteria:**
  - **Given** a valid, unexpired token in the URL (`/register?token=...`),
  - **When** the user provides their name and sets an 8+ character password,
  - **Then** the account is activated, the token is permanently consumed, and the user enters the Interviewer Workspace without an intermediate login step.

Module B: Candidate Portal & Apply Wizard

FR-004: Multivariable Job Search & Filtering

- **Description:** Candidates can search active job requisitions by keyword, location, and work mode.
- **Linked Test Cases:** `TC-CAND-001`
- **Acceptance Criteria:**
  - **Given** a list of published jobs,
  - **When** a candidate types a keyword and selects filter chips,
  - **Then** the listing updates in real time with the matching job count.

FR-005: Dynamic Application Form Rendering

- **Description:** Candidate application wizard dynamically renders input fields according to the employer's per-job form schema.
- **Linked Test Cases:** `TC-CAND-002`
- **Acceptance Criteria:**
  - **Given** a job with custom sections and fields (Text, Select, Number, File Upload),
  - **When** a candidate opens the apply wizard,
  - **Then** each field is rendered according to its data type with required-field indicators.

FR-006: Local Draft Persistence & Unsaved Changes Guard

- **Description:** Candidate form responses must be saved locally to prevent data loss.
- **Linked Test Cases:** `TC-CAND-003`
- **Acceptance Criteria:**
  - **Given** a candidate who has completed Step 1 and part of Step 2,

    - **When** the page is reloaded,
    - **Then** previously entered values are restored from browser local storage.
    - **When** the user attempts to close the tab with unsaved inputs,
    - **Then** a native `beforeunload` warning is triggered.

FR-007: Unauthenticated Guest Candidate Application

- **Description:** Candidates can apply without prior registration via guest submission.
- **Linked Test Cases:** `TC-CAND-006`
- **Acceptance Criteria:**
    - **Given** an unauthenticated visitor,
    - **When** they supply their name, email, and upload a resume via `/public/guest-resume`,
    - **Then** the backend links the application to an existing profile if matching email is found, or provisions a guest candidate record automatically.

Module C: Recruiter Portal & Pipeline Management

FR-008: Dynamic Application Form Builder

- **Description:** Recruiters can configure custom sections and custom questions for each job posting.
- **Linked Test Cases:** `TC-REC-001`
- **Acceptance Criteria:**
    - **Given** a recruiter editing a job,
    - **When** they add sections and questions with validation rules,
    - **Then** the schema is persisted in JSONB format and enforced on subsequent candidate applications.

FR-009: AI Resume Match Scoring (`aiScore`)

- **Description:** Automatically scores candidate resumes against job requirements (0–100%).
- **Linked Test Cases:** `TC-REC-002`
- **Acceptance Criteria:**
    - **Given** a candidate application with a resume,
    - **When** the application is submitted,
    - **Then** the backend calculates an `aiScore` (0-100%) comparing extracted resume text with required job skills, falling back to local TF-IDF if Groq LLM API is unset.

FR-010: Strict Forward-Only Pipeline Stage Guard

- **Description:** Prohibits reverting candidate cards back to the initial `APPLIED` status.
- **Linked Test Cases:** `TC-SM-001`

- **Acceptance Criteria:**
  - **Given** an application in `SCREENING`, `INTERVIEW`, or later,
  - **When** a user attempts to update status to `APPLIED`,
  - **Then** the request is rejected with `400 Bad Request` ("Cannot revert candidate to APPLIED stage").

FR-011: Open Interview Hard Block

- **Description:** Blocks advancing to `OFFER_STAGE` while an interview round is open.
- **Linked Test Cases:** `TC-SM-002`
- **Acceptance Criteria:**
  - **Given** an application with an interview in `ASSIGNED` or `CONFIRMED` status,
  - **When** the recruiter attempts to transition the candidate to `OFFER_STAGE`,
  - **Then** the transition is blocked: "Open interviews must be completed or cancelled before advancing."

Module D: Interviewer Workspace & MeetX Scheduling

FR-012: Recurring Weekly Availability Configuration

- **Description:** Interviewers can specify recurring weekly time windows when they are available for interviews.
- **Linked Test Cases:** `TC-INT-001`
- **Acceptance Criteria:**
  - **Given** an interviewer in the Interviewer Portal,
  - **When** they configure recurring days and times (e.g. Mon/Wed 14:00–18:00) with a timezone,
  - **Then** the availability schedule persists and is used by the slot generation engine.

FR-013: Self-Service Slot Booking with Atomic Concurrency Guard

- **Description:** Candidates choose an interview slot; concurrent double-booking is blocked.
- **Linked Test Cases:** `TC-CAND-004`, `TC-CAND-005`
- **Acceptance Criteria:**
  - **Given** an open interview slot,
  - **When** a candidate confirms booking,
  - **Then** the slot updates to `BOOKED` and the interview updates to `CONFIRMED`.
  - **When** a second candidate attempts to book the same slot concurrently,
  - **Then** the second request fails with `409 Conflict`.

FR-014: MeetX Video Room Generation & URL Privilege Split

- **Description:** Schedular (MeetX) creates video rooms with separate host and candidate join URLs.
- **Linked Test Cases:** `TC-MEET-001`, `TC-MEET-002`
- **Acceptance Criteria:**
  - **Given** a confirmed interview booking,
  - **When** the meeting provider is invoked,
  - **Then** it records `hostJoinUrl` (privileged controls for interviewer) and `candidateJoinUrl` (attendee link for candidate).

FR-015: Structured Scorecard & Cultural Fit Submission

- **Description:** Interviewers submit rubric criteria scores (1–5), Cultural Fit Score, and recommendation.
- **Linked Test Cases:** `TC-INT-003`, `TC-INT-004`
- **Acceptance Criteria:**
  - **Given** a completed interview call,
  - **When** the interviewer submits evaluation ratings, Cultural Fit Score (1-5), and recommendation (`STRONG_HIRE`, `HIRE`, `LEANING_HIRE`, `NO_HIRE`),
  - **Then** the feedback is recorded and the interview status updates to `COMPLETED`.

Module E: Offer Letters & Document Verification

FR-016: Branded Offer Letter Generation & Variable Interpolation

- **Description:** Generates PDF offer letters with interpolated candidate and compensation placeholders.
- **Linked Test Cases:** `TC-OFF-001`
- **Acceptance Criteria:**
  - **Given** a candidate in `OFFER_STAGE`,
  - **When** a recruiter generates an offer,
  - **Then** placeholders (`{{candidate_name}}`, `{{salary}}`, `{{start_date}}`, `{{expiry_date}}`) are replaced with localized values, a PDF is created, and uploaded to Cloudflare R2.

FR-017: Stale Offer Auto-Expiration Cron

- **Description:** Hourly background job expires stale extended offers past their expiration timestamp.
- **Linked Test Cases:** `TC-OFF-002`
- **Acceptance Criteria:**
  - **Given** an offer in `EXTENDED` status with `expiresAt < NOW()`,
  - **When** the hourly cron runs,
  - **Then** the offer status updates to `EXPIRED` and the candidate cannot accept the offer.

FR-018: Multi-File Compliance Document Upload

- **Description:** Candidates in `DOCUMENT_VERIFICATION` can upload multiple files per document requirement.
- **Linked Test Cases:** `TC-DOC-001`
- **Acceptance Criteria:**
  - **Given** an application in `DOCUMENT_VERIFICATION`,
  - **When** a candidate uploads multiple files for a document (e.g., front/back ID),
  - **Then** files are persisted to Cloudflare R2 and attached as `DocumentFile` records in `PENDING` state.

FR-019: Document Rejection & Re-Upload Loop

- **Description:** Rejecting a document requires a mandatory reason; candidate re-upload clears audit stamps.
- **Linked Test Cases:** `TC-DOC-002`
- **Acceptance Criteria:**
  - **Given** a pending document,
  - **When** a recruiter rejects it with a reason ("Scan illegible"),
  - **Then** the document status updates to `REJECTED`.
  - **When** the candidate re-uploads a replacement file,
  - **Then** `rejectionReason`, `verifiedBy`, and `verifiedAt` are cleared and status resets to `PENDING`.

FR-020: Finalize Hire Gate

- **Description:** Moving to `HIRED` is locked until all required documents are verified.
- **Linked Test Cases:** `TC-SM-004`
- **Acceptance Criteria:**
  - **Given** an applicant in `DOCUMENT_VERIFICATION`,
  - **When** any mandatory document is missing or not in `VERIFIED` status,
  - **Then** the "Finalize Hire" button is disabled and status dropdown cannot be set to `HIRED`.

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7. Non-Functional Requirements (NFR)

ID	Category	Requirement Specification
NFR-001	**Multi-Tenancy**	Every company-specific table query must include `companyId` filter. Cross-tenant reads or writes are strictly prohibited.

ID	Category	Requirement Specification
NFR-002	**Session Security**	Session cookies must be HTTP-only, secure in production, with `SameSite=Lax`. Cross-portal cookie collision is mitigated via ADR-001.
NFR-003	**Webhook Authenticity**	All incoming Schedular/MeetX webhooks must validate HMAC SHA-256 signatures before processing payloads (fail-closed).
NFR-004	**Performance**	Dynamic application wizard steps and dashboard pages must render in under 800ms. Search results return within 200ms.
NFR-005	**Storage Integrity**	Resumes, offer letters, and compliance files must be stored on Cloudflare R2 with cascade protection (`SetNull` on application delete).
NFR-006	**Availability & Resilience**	System must gracefully degrade to mock meeting links and local TF-IDF scoring if external third-party APIs (Schedular, Groq) are unreachable.
NFR-007	**Auditability**	Every document verification, rejection, and pipeline stage advancement must record timestamps and user IDs.

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8. Open Questions & Future Enhancements

1. **Vendor Job Board Direct APIs:** Currently, XML feeds are active; direct API posting will be enabled once vendor partner keys (LinkedIn/Naukri enterprise credentials) are acquired.
2. **Automated Reminders:** Adding automated SMS/WhatsApp reminders for upcoming interview calls 1 hour prior to session start.
3. **Custom Domain SSL Automation:** Automated provisioning of Let's Encrypt certificates for company CNAME domains.